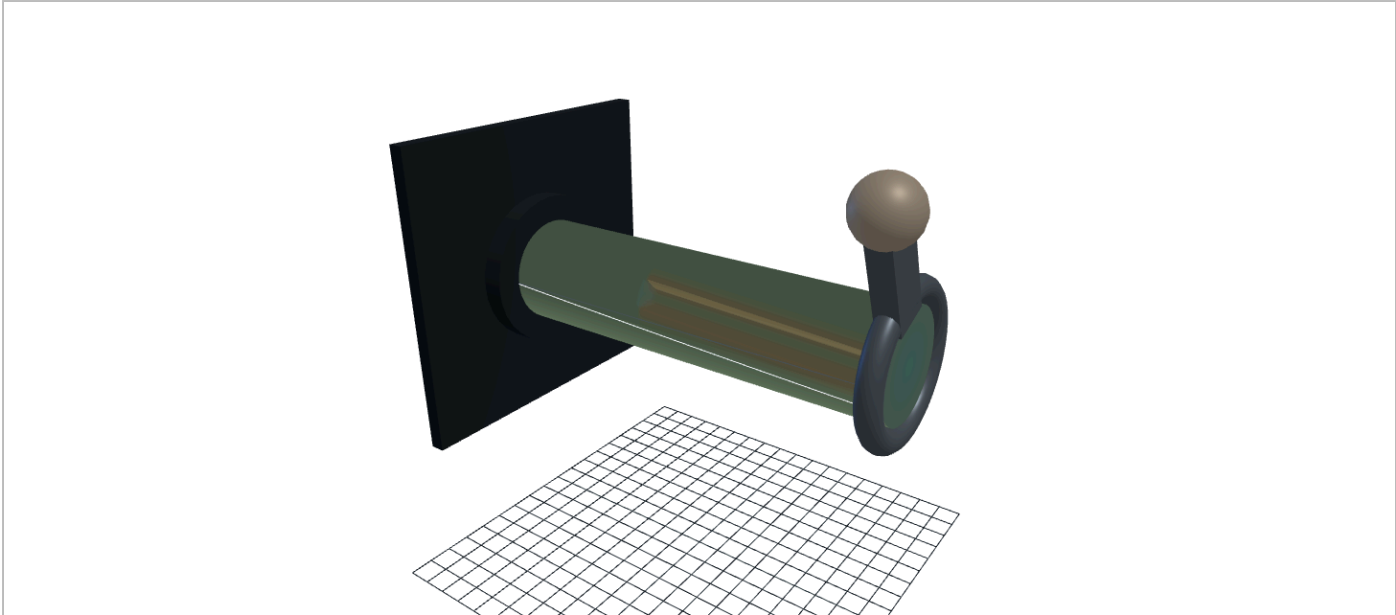


Shaft in Torsion — design calculation

Ø25 solid × 100 mm · Steel 1045 (cold drawn) · End-milled keyseat (profiled) r 0.50 mm (r/d 0.020)
T = 120.0 N·m at 1450 rpm = 18.22 kW · MechCalc – design check, not a substitute for full analysis

n = 2.61

τ_{max} 117.3 MPa vs 306 MPa allowable · capacity 312.7 N·m · twist 0.23° (limit 0.20°)



The shaft as modelled, shaded by shear stress against 0.577·σ_y. The hot band sits at the end-milled keyseat (profiled); the end face carries the radial τ ∝ r gradient. Twist shown ×102.

Section & loading

Polar second moment J	38350 mm ⁴
Section modulus Z _p	3068 mm ³
Shear modulus G	79.5 GPa
Nominal surface shear	39.1 MPa
Peak shear, K _{ts} 3.00	117.3 MPa
Torsional stiffness	531.8 N·m/°

Equations used

τ = T·c / J = 16T / πd³ – Surface shear stress (solid shaft)
J = π(d⁴ – d_i⁴) / 32 – Polar second moment – hollow or solid
τ_{max} = K_{ts} · T·c / J – Keyseat, fillet or groove multiplies it
K_{ts} = K_{ref}·(r/d ÷ r_{ref})^{−0.238} – ...and the radius you cut sets K_{ts}
τ_{allow} = 0.577 · σ_y – Distortion-energy shear yield
θ = T·L / G·J, G = E / 2(1+ν) – Angle of twist and the modulus it works on
k_t = G·J / L – Torsional stiffness, N·m per radian
T = 9549 · P[kW] / n[rpm] – Power ≠ torque at shaft speed
n = τ_{allow} / τ_{max} – Safety factor against shear yield

Worked calculation

1 · THE SECTION

Torsion of a circular shaft is the one case where the elementary theory is exact: plane sections stay plane and round, so each cross-section simply rotates. Everything follows from the polar second moment.

POLAR SECOND MOMENT

J = π(d^4 - d_i^4) / 32 = π(25.0^4 - 0) / 32 = 38350 mm^4

Section modulus Z_p = J/c = 3068 mm^3, with c = d/2 = 12.50 mm.

Solid section. A Ø25 shaft bored to Ø15 would keep 87% of this J for 64% of the weight — worth a thought if mass matters.

2 · THE STRESS THAT MATTERS IS LOCAL

NOMINAL SURFACE SHEAR

τ = T·c / J = 120.0 × 0.0125 / 3.835e-8 = 39.1 MPa

Zero on the axis, linear with radius, maximum at the surface — which is why only the outer fibres are doing real work.

PEAK SHEAR AT THE FEATURE

τ_max = K_ts·τ = 3.00 × 39.1 = 117.3 MPa

End-milled keyseat (profiled), r = 0.50 mm on Ø25.0 → r/d = 0.020.

Where that K_ts comes from. A concentration factor is not a property of the feature, it is a property of how sharp the feature is — a table entry has to name an r/d to mean anything. Shigley Table 7-1 gives a shoulder fillet in torsion at two radii, K_ts 2.2 at r/d = 0.02 and 1.5 at r/d = 0.1. Two points fix a power law:

K_ts(r/d) = K_ref · (r/d ÷ 0.020)^-0.238

Exact through both fillet anchors; for the keyseat and groove it is an interpolation anchored on their handbook figure at the standard radius, borrowing the fillet's curve shape on the grounds that all of them are the same physics — a notch in torsion. Honest between r/d 0.01 and 0.3.

3 · THE CHECK

Pure torsion is pure shear, and the distortion-energy criterion puts shear yield at 0.577·σ_y — the single most useful number on this page. A shaft that is comfortable in tension gives way at a little over half that stress in torsion.

QUANTITY	VALUE
Material	Steel 1045 (cold drawn)
Shear modulus G = E / 2(1+ν)	79.5 GPa
Allowable shear 0.577·σ_y	306 MPa
Peak shear at 120.0 N·m	117.3 MPa
Safety factor n = τ_allow / τ_max	2.61
Torque at n = 1	312.7 N·m
Power at 1450 rpm	18.22 kW

At 120.0 N·m this shaft holds with margin — n = 2.61. Without the keyseat the same shaft would read n = 7.82 — the feature is costing you a factor of 3.00.

4 · WIND-UP

ANGLE OF TWIST

θ = T·L / G·J = 120.0 × 0.100 / (79.5e9 × 3.835e-8) = 0.23°

2.26°/m · torsional stiffness GJ/L = 531.8 N·m per degree.

Strength is rarely the only limit. Wind-up puts a driveshaft's timing out, loses a leadscrew its position, and stores energy that comes back as torsional vibration. The workshop rule of thumb is **1° per 20 diameters** of length — here 0.20°, and you are at **0.23°**.

Over the wind-up rule by 1.13×, even though the stress check passes. Long slender shafts fail on stiffness long before they get near shear yield, and no amount of better steel fixes it — G barely varies between steels. Diameter does: J goes as d^4 , so 20% more diameter halves the twist.

SCOPE

Static torque only, on a prismatic circular section of isotropic material. **No bending, no axial load, no combined-stress envelope, and no fatigue.** A real drive shaft sees all of them: a rotating shaft carrying a steady bending load is in fully reversed bending, where the fatigue limit and the fatigue notch factor K_f — not K_{ts} — govern. Treat this as the first sizing pass and apply a service factor for shock loads. A keyway also removes section, which is folded into K_{ts} here rather than into J. Non-circular sections warp out of plane and obey none of this.

REFERENCES

Budynas & Nisbett, **Shigley's Mechanical Engineering Design** – torsion, ch. 3; K_t estimates, Table 7-1

Pilkey, **Peterson's Stress Concentration Factors** – shafts in torsion, keyseats and shoulder fillets

ANSI B17.1 – keys and keyseats, standard proportions $w = d/4$, $h = d/8$

Design tips

1 • Size the feature, not the section

The nominal stress almost never governs — the keyseat does. A profiled keyway triples the surface shear at its bottom corners, so a shaft with a comfortable safety factor on plain section is at yield the moment you cut the keyway for the pulley. **Run the check at the feature and nowhere else.**

Yours: plain section would be $n = 7.82$, at the feature it is $n = 2.61 - K_{ts} 3.00$.

2 • The radius is the cheapest strength you can buy

Doubling a fillet radius costs a different tool and nothing else, and it buys back real capacity. Going the other way — letting the shop leave a sharp corner because the drawing didn't say — quietly spends it. **Put the radius on the drawing and inspect it.**

Yours: $r = 0.50 \text{ mm}$ ($r/d 0.020$) $\rightarrow K_{ts} 3.00$. At double the radius K_{ts} would be **2.54**, worth **18%** more torque.

3 • Diameter beats material, every time

Torsional capacity goes as d^3 and stiffness as d^4 , while swapping steel for better steel changes σ_y and leaves G essentially alone. **A 26% bigger shaft is twice as strong and 2.5x as stiff** — reach for size before you reach for an alloy, unless weight is the binding constraint.

Yours: $\varnothing 25.0 \rightarrow \varnothing 31.5$ would take **625 N·m** instead of 313, and twist 0.09° instead of 0.23° .

4 • Bore the middle out before you shave the outside

Stress grows with radius, so the metal near the axis carries almost nothing and weighs just as much as the metal at the surface. **A tube beats a solid bar of the same mass every time.** It is also why the winner in bending loses in torsion: an open section like an I-beam has no closed shear path, and a slit tube is orders of magnitude softer in twist than the same tube intact.

A $\varnothing 15$ bore here would keep **87%** of J for **64%** of the metal.

5 • Check the wind-up before you sign off

Long shafts fail the twist limit long before the stress limit, and the failure is not dramatic — it is a machine that loses its timing, a leadscrew that lags its command, a drivetrain that rings. **1° per 20 diameters** is the usual ceiling; instrument drives and indexing shafts want far less.

Yours: **0.23°** over 100 mm, limit 0.20° — **1.13x over**.

6 • A rotating shaft is a fatigue problem, not a static one

This page checks static yield. Put a pulley on that shaft and the belt tension bends it; spin it, and every fibre goes through fully reversed bending once per revolution. **That is where shafts actually break** — at the keyway, in fatigue, at a stress the static check called comfortable. Size statically here, then run the fatigue check with K_f , surface finish and size factors before it turns.

Static torque only — no bending, no combined stress, no fatigue. K_{ts} is a first-iteration estimate interpolated from Shigley Table 7-1 anchors, not a Peterson chart lookup. Material values are typical reference figures; verify before production use.